

SMART WASTE BIN: AN IOT BASED SOLUTION FOR CLASSIFICATION AND MONITORING

ABSTRACT

The rapid growth of urban populations has led to increased waste generation, putting pressure on traditional waste management systems. Overflowing bins, inefficient collection routines, and poor segregation are common challenges in both urban and residential settings.

This project presents a **cost-effective, IoT-based Smart Bin system** designed to improve local waste handling through **real-time monitoring and basic waste segregation**. The system uses **ultrasonic sensors** to measure the fill level of two separate bins (wet and dry), a **moisture sensor** to detect the type of waste, and an **Proximity sensor** to identify the presence of waste near the bin opening. A **servo motor** then automatically directs the waste to the appropriate bin based on its moisture content.

The system is controlled by an **ESP32 microcontroller** and features a **web-based dashboard** hosted locally, providing live updates on bin status and waste type. The dashboard supports **multilingual display** in English, Hindi, and Kannada, making it accessible to a wider audience.

While simple in design, the Smart Bin achieves its goal of automating waste segregation and providing fill-level feedback in a user-friendly format. Its **modular, low-cost architecture** makes it suitable for deployment in campuses, apartments, and small communities, offering a scalable solution for smarter and more hygienic waste disposal.

PROBLEM STATEMENT

With rapid urbanization and population growth, effective waste management has become a major challenge for municipal authorities and communities. Traditional waste collection methods rely on fixed schedules without considering the actual fill levels of garbage bins. This often results in two major issues: overflowing bins that cause health hazards and environmental pollution, and inefficient collection routes that waste time, fuel, and manpower.

Moreover, the lack of real-time data makes it difficult for waste management departments to monitor and manage bin status across multiple locations. Manual inspection of each bin is time-consuming, labour-intensive, and impractical for large-scale implementation. These inefficiencies lead to increased operational costs, citizen dissatisfaction, and an overall decline in cleanliness and hygiene in urban and rural areas alike.

Therefore, there is a pressing need for a smart and automated solution that can monitor the status of waste bins in real-time, send timely alerts, and assist in planning efficient collection routes. This project aims to address this problem by developing an IoT-based Smart Bin system that offers a practical, scalable, and cost-effective approach to modern waste management.

SOFTWARE AND HARDWARE REQUIREMENTS

Table 1.1-Hardware Components:

COMPONENTS	DESCRIPTION
ESP 32 Dev Module	Microcontroller with Wi-Fi capabilities
Ultrasonic sensor x2	Accurate fill level detection
Proximity Sensor	Detects presence /motion of objects
Arduino Uno	For voltage supply
Moisture sensor	Detects moisture level
Servo motor	Control angular, linear position and acceleration
Jumper Wires	For electrical connections

• **Table 1.2- Software Tools:**

SOFTWARE	PURPOSE
Arduino IDE	Code development and uploading to ESP32
ESP32 Libraries	Used for Wi-Fi connectivity, Servo control, and sensor integration

DESIGN METHODOLOGY

Figure 1.1-Block Diagram:

The block diagram illustrates the functional flow of the Smart Waste Bin. It includes two sensors—moisture and Proximity sensor—connected to an ESP32 microcontroller. The ESP32 uses Wi-Fi to upload data to the cloud, which is then viewed on a web-based dashboard.

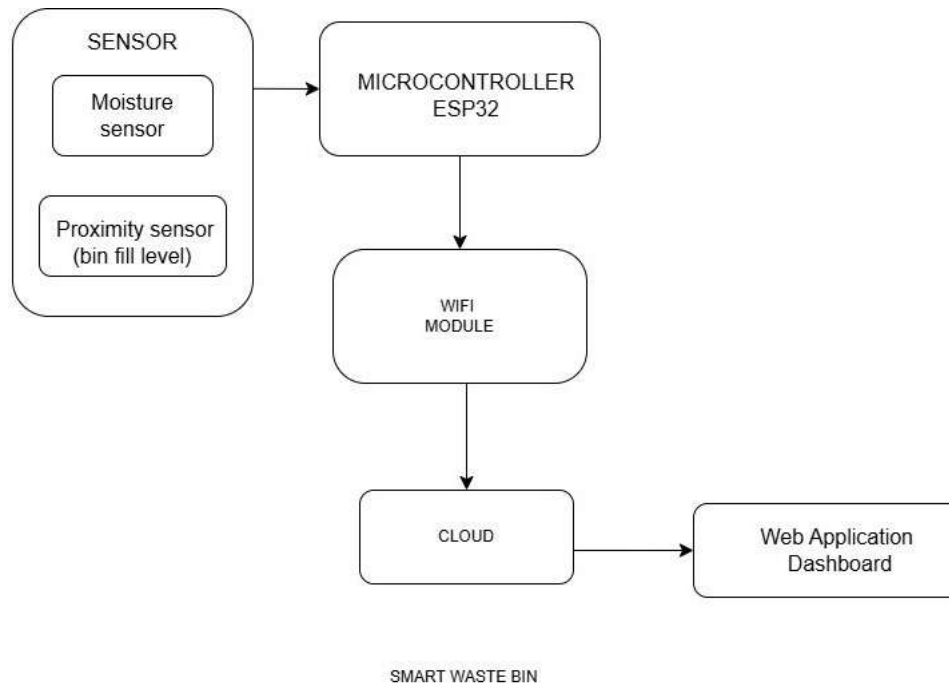


Figure 1.2- Circuit Diagram:

This diagram presents the wiring connections between the ESP32, Proximity sensor, moisture sensor, servo motor, and ultrasonic sensors. The ESP32 controls all components, while Arduino Uno is used only for providing 5V power supply, since it was already available.

Note:

- Breadboard was not used in the final implementation.
- Power supply (3.3V/5V) was taken from both ESP32 and Arduino Uno.
- The ESP32 handled data logic, and the Arduino only provided regulated power.

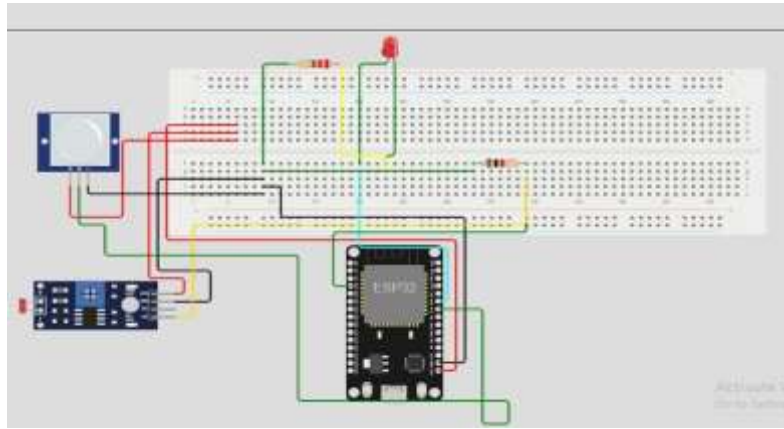
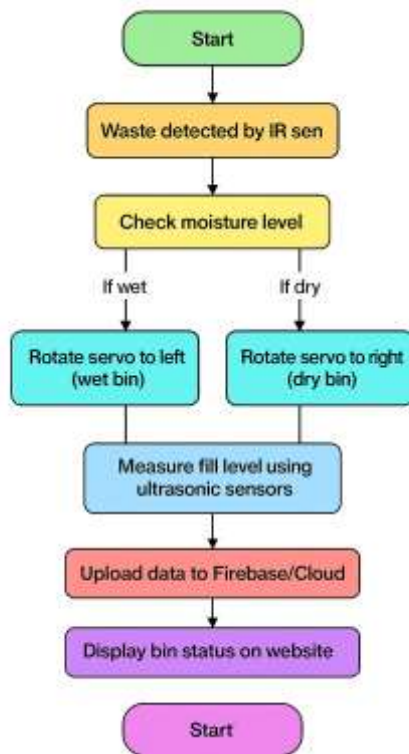


Figure 1.3: Flowchart of Working



RESULTS

The implementation of the Smart Waste Bin Management system using IoT components successfully demonstrated the feasibility of real-time waste monitoring and automation in a cost-effective and efficient manner. The ultrasonic sensor accurately measured the fill levels of the bin and transmitted the data to the ESP32 microcontroller, which in turn could be configured to send alerts when the bin approached its maximum capacity.

The Proximity sensor effectively detected the presence of a person near the bin, allowing the servo motor to open the lid automatically, thus minimizing physical contact and promoting hygiene. Additionally, the moisture sensor provided basic differentiation between wet and dry waste, which can be useful for future segregation mechanisms.

All components worked in coordination, and the system was able to provide timely alerts and responses, validating the concept of smart waste management. The results confirmed that such a system could be scaled and customized for use in urban residential areas, campuses, office spaces, and smart city infrastructure to improve the overall efficiency of waste collection and sanitation.

In conclusion, the project achieved its objective of designing a prototype that automates waste monitoring and offers intelligent features to support cleaner and smarter waste handling practices. Future enhancements may include integration with a cloud dashboard, solar power support, or AI-based waste analytics for large-scale deployments.

Table 1.3: Test Case Summary

Test Case	Condition	Expected Output	Observed Output	Status
TC1	Waste is wet	Waste is put into Left side of bin	Wet waste in wet bin	Pass
TC2	Waste is dry	Waste is put into right side of bin	Dry waste in dry bin	Pass

Future Scope

1. Cloud Integration and Mobile App

Integrating the system with cloud platforms can enable centralized data storage, advanced analytics, and remote access through a web or mobile application. This would allow municipal authorities or users to monitor multiple bins across different locations in real time.

2. Route Optimization Using GPS

By adding GPS modules and integrating with route planning algorithms, the system can help optimize garbage truck routes based on bin fill levels, reducing fuel consumption and improving collection efficiency.

3. AI and Data Analytics

Implementing machine learning models can help in predicting waste generation patterns, identifying peak usage times, and automating decision-making for resource allocation.

4. Solar Power Integration

To enhance energy efficiency and allow deployment in remote areas, the system can be powered using solar panels. This would reduce dependency on traditional power sources and support sustainability goals.

5. Smart Waste Segregation

With the use of advanced sensors and image processing, the system can be upgraded to automatically identify and segregate waste as biodegradable, non-biodegradable, or recyclable, making waste disposal more environmentally friendly.